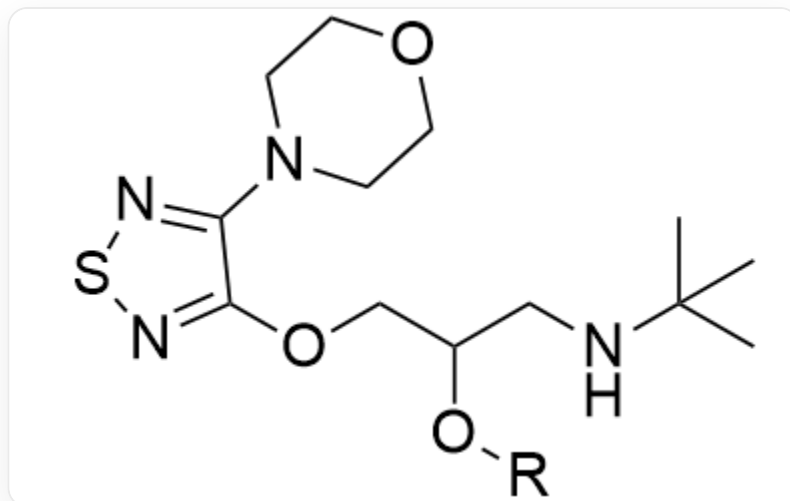


Question

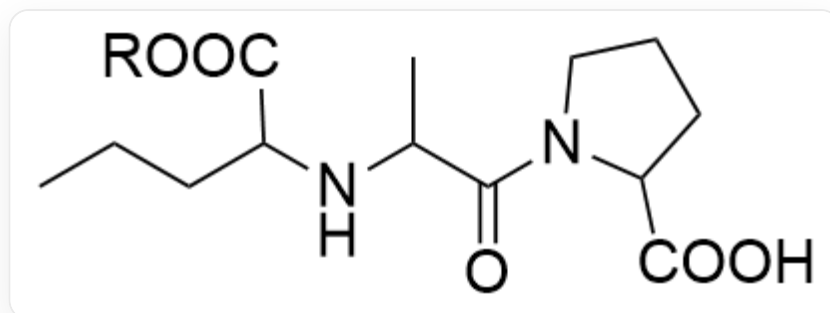
For the following drug molecule fragments, chemists are attempting to improve their properties in specific aspects by optimizing the side chain group *R*:

- Drug molecule 1: Increase lipophilicity



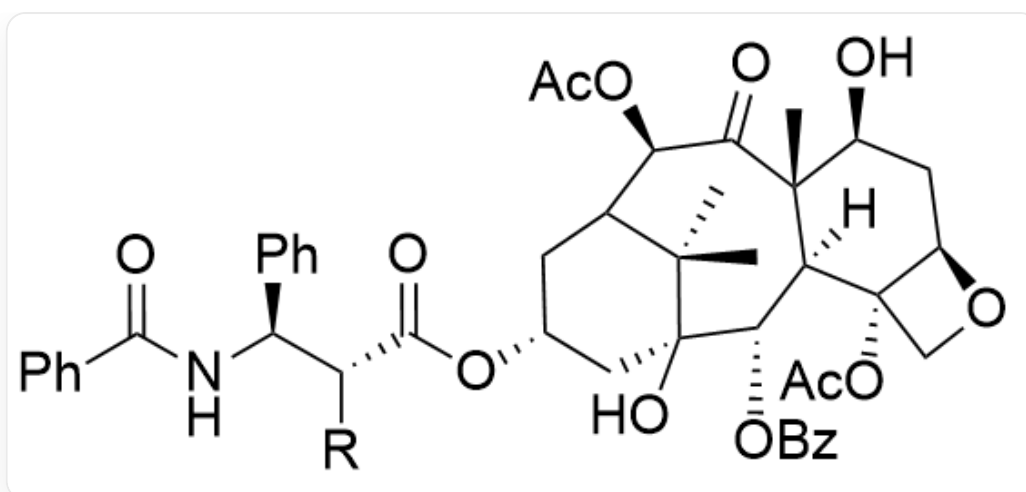
SMILES : CC(C)(NCC(O[R])COC1=NSN=C1N2CCOCC2)C

- Drug molecule 2: Increase lipophilicity



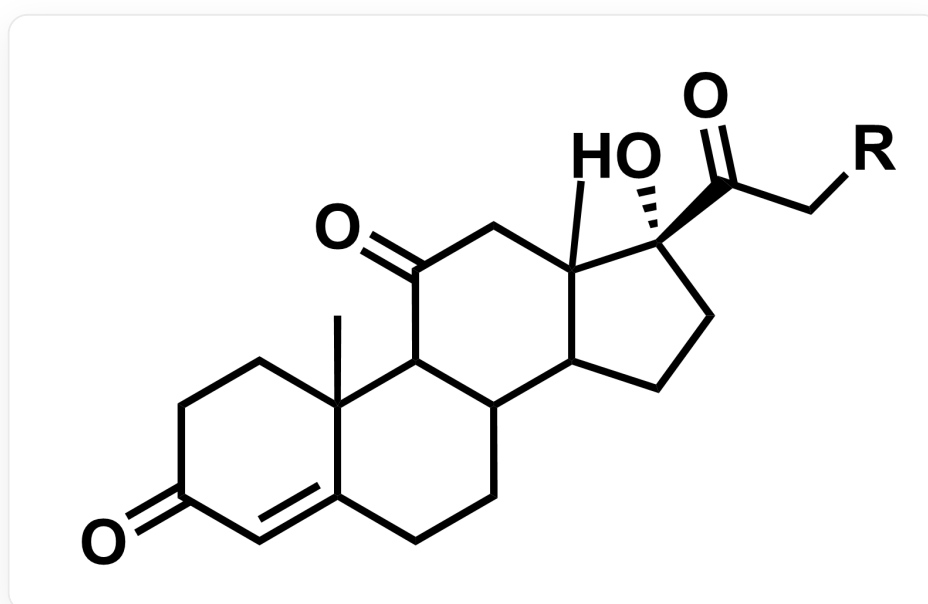
SMILES : CCCC(C(O[R])=O)NC(C)C(N1C(C(O)=O)CCC1)=O

- Drug molecule 3: Increase water solubility



SMILES : C[C@]12[C@@]([C@H](OC(C3=CC=CC=C3)=O)[C@]4(O)C[C@H](OC([C@@H]([C@H](C5=CC=CC=C5)NC(C6=CC=CC=C6)=O)[R])=O)CC(C4(C)C)[C@@H](OC(C)=O)C2=O)([H])[C@@](CO7)(OC(C)=O)[C@H]7C[C@@H]1O

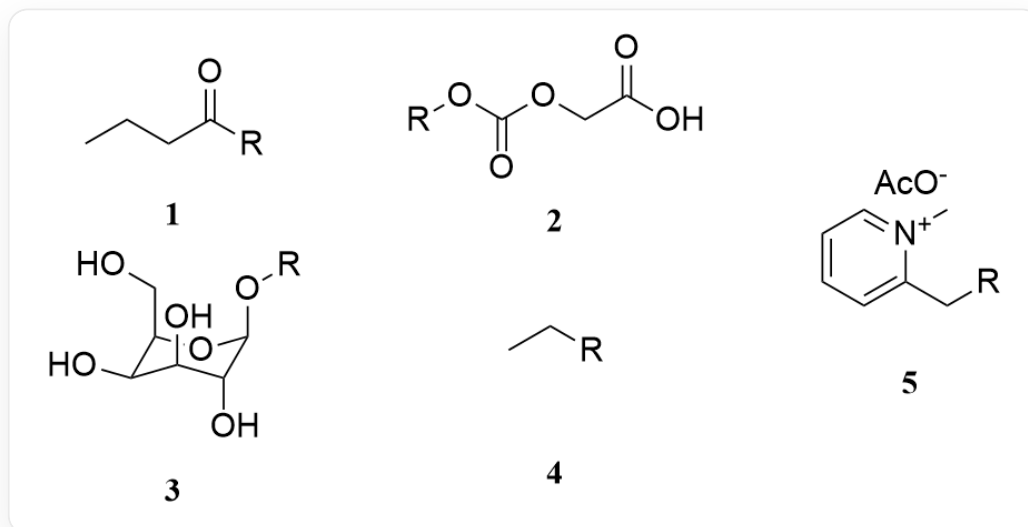
- Drug molecule 4: Increase specificity for the site of action:



CC12CCC(=O)C=C2CCC3C4CC[C@](C(=O)C[R])(C4(C)CC(=O)C31)O

From the following numbered side chain groups, find the groups that can achieve specific property improvements (each side chain group corresponds to one and only one drug molecule fragment, and one drug

molecule fragment may correspond to multiple side chain groups). The number of each side chain group is marked below, where R represents the position connected to the drug molecule fragment:



SMILES从1到5依此为: CCCC([R])=O [R]OC(OCC(O)=O)=O [O][C@H]1[C@@H](CO)O[C@@H](O[R])
[C@H](O)[C@H]1O CC[R] [R]CC1=CC=CC=[N+]1C CC(=O)[O-]

Let the sequence of drug molecule numbers be $a_{1\sim 4}$, and the sequence of side chain group numbers be $b_{1\sim 5}$. Calculate $x_{ij} = \frac{a_i}{b_j}$ for each pairing relationship, and further calculate the sum of all x_{ij} as z . Select the correct value of z .

A. 4.68

B. 4.77

C. 4.93

D. 5.05

E. 5.68

F. 6.53

G. 6.93

H. None of the above options are accurate

Answer

Correct Answer: C

Detailed Explanation

Match the drug molecular fragments with the side chain groups as follows:

Drug Molecules 1 & 2: Increase lipophilicity, typically achieved by introducing nonpolar hydrocarbon chains or esterifying to mask polar groups (e.g., -OH or -COOH).

CHECKPOINT

0.5 PTS

Increasing lipophilicity requires masking polar groups

Side chain 1 (butyryl) and side chain 4 (ethyl) are both nonpolar groups that effectively increase lipophilicity. Side chain 1 (butyryl) can acylate hydroxyl groups to form esters, while side chain 4 (ethyl) can esterify carboxyl groups to form esters.

CHECKPOINT

0.5 PTS

Side chain 1 (butyryl) and side chain 4 (ethyl) increase lipophilicity

The *R* group of Drug Molecule 2 is attached to a carboxyl group ($-\text{COOH}$). Converting it to an ethyl ester (attaching side chain 4) is a classic method in medicinal chemistry to enhance lipophilicity and oral bioavailability. Therefore, Molecule 2–Side Chain 4 is a reasonable match.

CHECKPOINT

1 PTS

Molecule 2 matches with Side Chain 4

The *R* group of Drug Molecule 1 is attached to a hydroxyl group (-OH). Esterification (attaching side chain 1) can also effectively mask polarity and increase lipophilicity. Therefore, Molecule 1–Side Chain 1 is a reasonable match.

CHECKPOINT

1 PTS

Molecule 1 matches with Side Chain 1

Drug Molecule 3: Increase water solubility. For a very large and highly lipophilic molecule (a paclitaxel analog), strongly polar, ionizable, or charged groups need to be introduced to significantly increase its water solubility.

Side chain 2 (containing carboxyl), side chain 3 (glucose), and side chain 5 (quaternary ammonium salt) are all polar or charged groups that increase water solubility. According to the given information, "each side chain group corresponds to and only corresponds to one drug molecular fragment, while a drug molecular fragment may correspond to multiple side chain groups." At least one of these three must correspond to Drug Molecule 4. Consider Drug Molecule 4 first.

Drug Molecule 4: Improve site-specificity. Enhance the drug's targeting to specific tissues or cells (e.g., tumor cells). A common strategy is to utilize transporters highly expressed on the surface of these cells, such as glucose transporters (GLUTs). Side chain 3 is a glucose group. Linking the drug to glucose enables its preferential uptake by cells with high GLUT expression, achieving targeted delivery.

CHECKPOINT

1 PTS

Side chain 3 (glucose) enables preferential uptake by cells with high GLUT expression

Thus, Molecule 4–Side Chain 3 is a standard strategy to achieve this goal.

CHECKPOINT

1 PTS

Molecule 4 matches with Side Chain 3

Therefore, to increase the water solubility of Drug Molecule 3, side chain 2 or 5 can be introduced.

CHECKPOINT

1 PTS

Molecule 3 matches with Side Chain 2 or Side Chain 5

Based on the above pairing relationships, calculate the $x_{ij} = a_i \times b_j$ values for each pair:

$$x_{1,1} = 1/1 = 1.00$$

$$x_{2,4} = 2/4 = 0.50$$

$$x_{3,2} = 3/2 = 1.50$$

$$x_{3,5} = 3/5 = 0.60$$

$$x_{4,3} = 4/3 = 1.33$$

CHECKPOINT

1 PTS

$$x_{1,1} = 1, x_{2,4} = 0.5, x_{3,2} = 1.5, x_{3,5} = 0.6, x_{4,3} = 1.33$$

Calculate the sum of $x_{i,j}$:

$z = \sum x_{i,j} = 4.93$, so select option C.

CHECKPOINT

1 PTS

$z = 4.93$, choose C